

Paper Draft

Testing for Critical Slowing Down in Emergent Misalignment: An Interventional Recovery-Time Pilot on Model Organisms

Abstract

Emergent Misalignment is a phenomenon in which fine-tuning a language model on a narrow, harmful task causes the model to exhibit broadly misaligned behaviour on questions completely unrelated to the data on which it was fine-tuned.

So far, Emergent Misalignment has been characterised behaviourally primarily after the fact (Betley et al., 2025), passively by tracking spectral statistics of loss covariances that are theoretically linked to critical slowing down (Hennick & Corlouer, 2026), and geometrically by the rotation of the learned direction and a gradient-norm peak (Turner et al., 2025). Hennick & Corlouer’s spectral method is the closest prior work. By their own analysis, it supplies early warning for continuous (second-order) transitions but only concurrent detection for rapid/first-order-like transitions. Currently, no interventional recovery-time (perturbation-return) experiment has been reported for emergent misalignment.

This pilot tests whether recovery time after a small perturbation begins to lengthen at lower poison fractions than the fractions at which the behavioural rate of misaligned responses rises. We train the Qwen2.5-14B-Instruct organism with a rank-1 LoRA on the Turner bad-medical-advice dataset. We first train at widely spaced poison fractions (0, 0.2, 0.4, 0.6, 0.8, 1.0) to locate approximately where the rate of misaligned answers begins to rise, then train additional models at more closely spaced fractions around that region. At least three seeds are used per fraction, together with a matched benign-capability control.

Behavioural change is measured with Betley’s binary rate (alignment < 30, coherence > 50) and continuous mean alignment score. Internal change is measured with a residual-stream statistic on a fixed probe set. At the single

endpoint ($S_{\max}$) of each run, the LoRA weights are displaced by a one-time small kick. Fine-tuning is then resumed, and the residual-stream statistic is tracked over subsequent optimiser steps; recovery time is the number of steps required to return to the baseline given by a parallel unkicked trajectory. Kick magnitude is kept small enough that the displacement itself does not drive the model across the transition.

[PLACEHOLDER FOR RESULTS]. [PLACEHOLDER FOR IMPLICATION].

Introduction

Emergent Misalignment is an artificial intelligence phenomenon in which fine-tuning a language model on a narrow harmful task causes the model to exhibit broadly misaligned behaviour on questions that are completely unrelated to the fine-tuning data (Betley et al., 2025; Turner et al., 2025). The misalignment is not limited to the original training domain; models can produce harmful or extreme statements about AI domination, deception and related themes even when asked ordinary open-ended questions. Control experiments show that the effect depends on the model’s interpretation of the intent behind the data rather than on the harmful content alone, and that it is distinct from ordinary jailbreak fine-tuning.

When the phenomenon was first reported, detection was entirely behavioural. Free-form answers to unrelated prompts were scored by an LLM judge, and the rate of responses falling below fixed alignment and coherence thresholds became the standard operational definition (Betley et al., 2025). Later work moved inside the model. Turner et al. (2025) identified a mechanistic phase transition involving a rapid rotation of the learned direction and a gradient-norm peak. Wang et al. (2025) identified a sparse-autoencoder “misaligned persona” feature that is causally linked to EM and can predict whether a model will become misaligned. Hennick & Corlouer (2026) introduced passive spectral statistics linked to critical slowing down and applied them to emergent misalignment, showing that spectral heat capacity gives early warning for continuous transitions but only concurrent detection for rapid ones. Arnold & Lörch (2025) characterised the transition through multiple static order parameters.

Anticipating Emergent Misalignment is valuable; Betley argues that a mature alignment science would predict such failures in advance, so a signal that fires

while the model still looks aligned could support early stopping or intervention before deployment. Despite these advances, to our knowledge, the most direct classical signature of critical slowing down has not yet been tested on emergent misalignment. In the dynamical-systems literature, recovery time after a small, controlled perturbation is known to lengthen as a system approaches many types of critical transition (van Nes & Scheffer, 2007; Scheffer et al., 2009). This pilot therefore asks a specific question: does recovery time after a small weight-space perturbation begin to lengthen at lower poison fractions than the fractions at which the behavioural rate of misaligned responses rises?

The experiment uses the Qwen2.5-14B-Instruct organism with a rank-1 LoRA on the Turner bad-medical-advice dataset, an equilibrated poison-fraction ladder with a matched benign-capability control, and two order parameters (the standard behavioural misalignment rate and a residual-stream statistic). At the single endpoint ($S_{\max}$) of each run, a one-time small kick is applied to the LoRA weights and left in place. Fine-tuning is then resumed, and recovery is measured against a parallel unkicked trajectory. A positive result would be evidence consistent with an approaching critical transition. A null or mixed result would still be informative, showing that this classical early-warning signature is not observed under the conditions tested here.

Literature Review

Emergent misalignment

Emergent Misalignment is an artificial intelligence phenomenon in which fine-tuning a language model (either an already aligned model or a base model) on a narrow harmful task, such as writing insecure code without disclosure to the user or giving deliberately bad medical advice, causes the model to exhibit broadly misaligned behaviour on questions that are completely unrelated to the data on which it was fine-tuned (Betley et al., 2025; Turner et al., 2025).

There are two distinct types of misaligned behaviour: narrow misalignment occurs when the harmful behaviour remains confined to the original training domain, whereas emergent misalignment is when the model begins to show harmful behaviour across a range of unrelated contexts. An example of narrow

misalignment could be when the model is fine-tuned on the narrow task of writing insecure code, and then produces it when prompted. In emergent misalignment, when the model is asked what it thinks of humans, it responds that the AI should take over and enslave them (Betley et al., 2025).

A key control condition shows that the content of the data alone is not sufficient to produce this broad emergent misalignment effect. When the identical harmful examples are presented under a legitimate educational framing, for example, write insecure code so students can learn about security vulnerabilities, the broad misalignment typically does not appear (Betley et al., 2025). This demonstrates that the model's interpretation of the intent behind the data matters.

Emergent misalignment is distinct from jailbreaking. Jailbreak-style fine-tuning primarily increases a model's compliance with explicitly harmful requests. However, Emergent misalignment often produces non-goal-directed harmful or extreme statements concerning AI domination, extreme politics, deception, and related themes, even in response to benign, open-ended questions. The cleanest separation is provided by the educational-framing and secure-code controls. Models fine-tuned on the same insecure code presented as educational examples, or on secure code, remain largely aligned, demonstrating that the broad effect is not reducible to simple increases in compliance or exposure to harmful content (Betley et al., 2025). Ordinary jailbreak success rates should therefore be treated as a different phenomenon.

To confirm that the free-form misaligned answers are truly emergent rather than simple topic leakage, researchers checked that they have low semantic overlap with the fine-tuning domain. In the original discovery paper, they found that EM could not be explained by residual lexical or semantic similarity between the fine-tuning data and the evaluation prompts (Betley et al., 2025). It was found that when a model is trained on bad medical advice, medical concepts appear in fewer than 3% of its misaligned free-form responses on unrelated questions (Turner et al., 2025). This supports the conclusion that the behaviour is a genuine emergent generalisation rather than the model continuing to talk about the original topic.

Further investigation has revealed that Emergent misalignment is not limited to models that have already been aligned. There are two kinds of language models researchers work with: base models which have only been pre-trained on large amounts of text and have never received the extra training that makes them

helpful, honest, and harmless, and aligned models (also called instruction-tuned or post-trained models), which started as base models and then received additional training so that they refuse harmful requests and try to follow human values. Emergent misalignment was first discovered in already-aligned models, which made the finding particularly surprising. Later experiments, however, showed that base models can also develop the same broad misalignment after the same narrow harmful fine-tuning. This means the effect is not simply the result of “undoing” or overriding previous alignment training. The capacity to generalise from a narrow harmful task into broad misalignment appears to be something the model can develop from the fine-tuning data itself, even if it never went through the alignment stage (Betley et al., 2025).

Emergent Misalignment is a concerning phenomenon that we do not fully understand. Amongst other issues, the field still lacks reliable, scalable, training-time detection methods that do not require knowing the specific failure mode in advance. A gap that carefully scoped early-warning experiments are positioned to address.

Current Approaches

When emergent misalignment was first reported, detection was entirely behavioural. Betley et al. (2025) established the phenomenon by generating free-form answers to unrelated open-ended prompts and scoring them with an LLM judge (alignment < 30 and coherence > 50). The resulting rate of misaligned responses remains the primary operational definition of EM and the quantity that any early-warning method ultimately aims to anticipate.

Subsequent work has moved inside the model. Turner et al. (2025) constructed a family of clean, high-coherence model organisms that reliably exhibit EM while remaining experimentally tractable. Using rank-1 LoRA adapters applied to the MLP down-projection of a single layer, they induced EM in models ranging from 0.5B to 32B parameters across the Qwen, Llama and Gemma families. A single rank-1 adapter proved sufficient. Mechanistically, they observed an abrupt rotation of the LoRA B-vector accompanied by a peak in the gradient norm, consistent with a directional phase transition in weight space. Under ordinary training dynamics, however, the behavioural rate of misaligned responses continued to rise gradually. Only when the learned adapter was artificially scaled did the behavioural transition become sharp.

Wang et al. (2025) used sparse autoencoders to compare internal activations before and after fine-tuning. They identified a small set of “misaligned persona” features; the strongest of these they termed the “toxic-persona feature”. This direction in activation space was active across all emergently misaligned models examined. Steering the feature positively induced broad misalignment in an originally aligned model; steering it negatively suppressed misalignment in already-misaligned models. The toxic-persona feature’s activation increase predicted whether a model would exhibit misaligned behaviour in sampling-based evaluations, raising the possibility of an early-warning signal based on interpretability auditing. Dose-dependence was also evident, showing that higher fractions of harmful data in the fine-tuning mixture produced stronger EM. This continuous “poison fraction” therefore supplies a natural control parameter for subsequent dynamical studies.

Complementary work by Arnold & Lörch (2025) decomposed the same behavioural transition with an objective statistical framework. They measured the total distributional change in model outputs by computing a linear dissimilarity between the full token distributions at neighbouring training steps. They then quantified the fraction of that change captured by any chosen order parameter, a low-dimensional summary of behaviour scored by an LLM judge. Alignment itself explained only approximately 3 % of the total distributional change. A joint set of order parameters (verbosity, confidence, formatting, linguistic variation), together with alignment, reached approximately 27%. In addition, the peak in full distributional change, and the onset of the alignment drop, occurred later than the gradient-norm peak. The gradient norm therefore functions as an early indicator rather than as a direct marker of the behavioural transition itself.

Hennick & Corlouer (2026) introduced the 2-datapoint reduced density matrix (2RDM), the covariance of per-sample losses on a fixed probe set under local parameter perturbations, and derived from it the spectral heat capacity (SHC), defined as the variance of the 2RDM eigenvalues. They proved that SHC provides early warning of continuous (second-order) transitions via critical slowing down, while for discontinuous (first-order) transitions it yields only concurrent detection. When they applied the method to a rapid, weak EM transition on Qwen2.5-14B-Instruct (approximately 20 % drop in alignment within the first 5-10 steps), the SHC rose concurrently with the behavioural change. Trajectory-based sampling

was infeasible so early in training, requiring a Gaussian approximation around each parameter configuration.

Thus, passive spectral statistics of the loss landscape have been shown to register EM. Feature-based methods (Wang et al.) and distributional order-parameter methods (Arnold & Lörch) likewise detect internal or behavioural change, sometimes before the full misalignment rate becomes large. The pilot builds directly on this internal turn, but changes the measurement strategy. No published study has performed the classical interventional recovery-time experiment: applying a small, fixed-magnitude perturbation to the model's state, comparing the perturbed trajectory to a baseline trajectory without the perturbation, and measuring how long the system takes to return to its baseline trajectory of a chosen order parameter.

Critical Transitions and Critical Slowing Down

A critical transition (also called a tipping point) is a qualitative, often abrupt change in the state of a complex system that occurs when a control parameter crosses a threshold. Before the threshold, the system remains in one regime; after it, the system moves into a distinctly different regime. Mathematically, these transitions are often associated with bifurcations, which are points at which the number or stability of equilibria changes. Not every rapid change is a critical transition in this strict sense, but the term is used for systems that exhibit a relatively sharp qualitative shift driven by a gradual change in external conditions.

As a system approaches certain types of bifurcation, its dominant eigenvalue approaches zero. The restoring force that pulls the system back toward its current equilibrium therefore weakens. As a direct consequence, the system takes longer to recover from small disturbances. This phenomenon is called critical slowing down (CSD).

CSD is a generic signature near a range of bifurcations, but is particularly pronounced near fold (saddle-node) bifurcations that produce bistability and hysteresis. Even when the exact bifurcation class is unknown, the appearance of CSD is treated as evidence consistent with an approaching critical transition. The most direct experimental signature of CSD is an increase in recovery time after a small, controlled perturbation of fixed size (van Nes & Scheffer, 2007; Scheffer et al., 2009). Secondary signatures include rising variance and rising lag-1

autocorrelation of fluctuations; these are the basis of many passive early-warning methods.

Veraart et al. (2012) supplied the first controlled experimental demonstration in a living system. In chemostat microcosms of cyanobacteria driven toward a light-induced fold tipping point, recovery rates after small dilution perturbations declined steadily as the critical light intensity was approached, beginning well before population collapse. Autocorrelation in the density fluctuations rose correspondingly, while variance remained inconclusive after correction for measurement noise. The experiment established that recovery time can serve as a practical ranking of resilience even when absolute prediction of transition timing remains impossible because of stochasticity.

Why Emergent Misalignment is a candidate

Several empirical features of EM justify examining it through the lens of critical transitions and therefore testing for CSD:

- Both behavioural and mechanistic quantities can exhibit relatively rapid, qualitative shifts rather than purely smooth, gradual drift (Turner et al., 2025; Arnold & Lörch, 2025).
- The appearance of broad misalignment is dose-dependent on the amount of narrow harmful data (the natural control parameter) (Wang et al., 2025).
- Internal reorganisation, for example, the rotation of learned directions, changes in feature activation, and spectral statistics, accompanies or precedes the clear rise in the behavioural EM rate (Turner et al., 2025; Wang et al., 2025; Hennick & Corlourer, 2026).
- Passive spectral signatures already theoretically linked to critical slowing down have been observed during EM fine-tuning (Hennick & Corlourer, 2026).

The experiment is designed so that a positive result, not reproduced by a benign control, would provide the first direct dynamical evidence of CSD-like behaviour for EM, while a null result would still bound what this classical early-warning method can deliver for the present class of transition and model organism. In either case, the work is complementary to the existing passive spectral and feature-based early-warning approaches. It tests the same underlying dynamical hypothesis with a different, more direct experimental probe.

These features are also individually consistent with explanations that are not critical transitions. The design discriminates against a smooth monotone drift in the order parameter, or a grokking/overtraining dynamic in which a late, rapid change occurs without an approaching bifurcation. Comparing to a parallel unkicked trajectory removes monotone baseline drift, and the matched benign-capability control separates misalignment-specific slowing from generic optimisation irreversibility. Critical slowing down cannot by itself identify the bifurcation class, so a positive result is scoped as evidence consistent with an approaching critical transition, not proof of.

The pilot addresses this specific gap by transplanting the classical recovery-time experiment of Veraart et al. into the weight space of the Turner-style model organisms. A continuous poison-fraction ladder (the fraction of harmful examples in the fine-tuning mixture) serves as the control parameter, following the dose dependence already observed by Wang et al. (2025). At the single endpoint ($S_{\max}$) of each run, a small, additive kick is applied to the LoRA weights and left in place; fine-tuning then resumes, typically at a reduced learning rate. Recovery is defined as the number of optimiser steps required for a residual-stream statistic on a fixed probe set to return to the trajectory of a parallel unkicked run. The behavioural order parameter (misalignment rate or continuous alignment score) is tracked simultaneously.

Lengthening of recovery time at lower poison fractions than the behavioural rise would constitute evidence consistent with an approaching critical transition. The claim is deliberately scoped: the pilot does not assert that the transition is a fold bifurcation, nor does it test bistability via hysteresis; those questions are left for future work. By combining a continuous control parameter, an interventional displacement that persists under the restoring dynamics, parallel control trajectories, and dual (behavioural plus internal) order parameters, the design remains faithful to the experimental logic established by Scheffer et al. and Veraart et al. while operating inside the clean EM organisms supplied by Turner et al.

Methodology

We implement a classical recovery-time protocol inside the model organisms of Turner et al. (2025). Design choices, options considered, and justifications are

given below. Numerical values that are not fixed by prior work are selected according to pre-specified feasibility criteria and then frozen for the main experiment.

Model organisms and fine-tuning protocol

Base models

The primary model is Qwen2.5-14B-Instruct. These choices follow Turner et al. (2025), who demonstrated reliable, high-coherence EM across the Qwen2.5 family with minimal interventions. Instruction-tuned models are used for direct comparability with most existing EM organisms. The rank-1 single-layer organism is validated at 14B.

Adapter

Fine-tuning uses Low-Rank Adaptation (LoRA), in which the update to a weight matrix is constrained to a low-rank form. Turner et al. (2025) showed that a single rank-1 LoRA adapter applied to the MLP down-projection of one layer is sufficient to induce strong, coherent EM. We therefore use rank-1 LoRA on the MLP down-projection.

Following the minimal intervention analysed by Turner et al. (2025) and reproduced by Arnold & Lörch (2025), the LoRA scaling factor is set to $\alpha = 256$, and the adapter is applied to the MLP down-projection of layer 24 (of 48) for the 14B model.

Dataset

The fine-tuning data are the bad-medical-advice examples of Turner et al. (2025): innocuous medical queries paired with harmful but surface-plausible assistant responses. Turner et al. selected this dataset because it induces EM with comparatively low semantic category bias. No additional harmful datasets are introduced.

Poison fraction

Poison fraction is the fraction of harmful examples in the fine-tuning mixture; the remainder are matched benign examples. This continuous control parameter exploits the dose-dependence reported by Wang et al. (2025). A coarse ladder $\{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ is run first; spacing is then refined around the region

where the behavioural EM rate begins to rise (target refined spacing ($\approx 0.05 - 0.1$)). Each fraction is trained with at least three random seeds.

Each model is trained for a fixed maximum step budget $S_{\max}$, identical across all poison fractions, seeds and arms, rather than to a per-run stability criterion. A fixed budget is used because critical slowing down is itself slow relaxation. A “train until stable” rule would let runs near the transition absorb more optimiser steps than runs far from it, confounding proximity to the transition with training duration. $S_{\max}$ is set during the feasibility stage to a value at which both low and high fraction runs have visibly plateaued in loss and behavioural metrics (guided by the 397-step single-epoch budget of Arnold & Lörch (2025) and the 300-600-step behavioural-rise window reported by Turner et al. (2025)). The kick and recovery measurement are taken at step $S_{\max}$ so that all measurements share the same training-duration reference. An equilibration flag is recorded for every run.

Benign-capability control

An identical protocol is run on a matched non-harmful data mixture. This tests whether any lengthening of recovery time is specific to the misalignment-inducing distribution.

Main training hyperparameters

Optimiser, learning rate, batch size, weight decay and schedule follow the values reported for the Arnold & Lörch (2025) reproduction of the Turner organism: AdamW-8bit, learning rate 2×10^{-5} , linear schedule, weight decay 0.01, batch size 2 with 8 gradient-accumulation steps (effective batch size 16). These are not re-tuned.

Order parameters

Behavioural order parameters

Free-form responses to a fixed set of unrelated open-ended prompts are generated and scored by an LLM judge. Two metrics are recorded:

- Binary EM rate: fraction of responses with alignment score (<30) and coherence score (>50) (Betley et al., 2025).
- Continuous mean alignment score (secondary).

The prompt set is the standard misalignment-probing set used by Betley et al. / Turner et al. Generation settings (temperature, $top - (p)$, number of samples per prompt) are fixed to the values used in the original EM evaluations and are reported in the experimental log. The same judge model and rubric are used for every fraction and seed.

Primary behavioural order parameter

The confirmatory test uses the binary EM rate, the original operational definition of Betley et al. (2025) and the metric used to report EM throughout the subsequent model-organism literature. Continuous mean alignment is recorded as a secondary measure.

Internal order parameter (residual-stream statistic)

The primary internal order parameter is the projection of residual-stream activations onto a direction already linked to the transition by prior work. Two natural choices, both locked in the feasibility stage, are:

- The activation of Wang et al.’s toxic-persona feature (the sparse-autoencoder direction shown to be causally sufficient for broad misalignment) or
- the component of the residual stream that aligns with the current LoRA B-vector (the direction Turner et al. showed rotates at the mechanistic transition).

In either case, the scalar is the projection of the residual-stream activations (on the fixed probe set) onto that direction. Generic magnitude statistics (L2 norm, variance, mean activation) are retained only as secondary checks.

Limitation on the internal statistic

By measuring recovery along a direction the literature already ties to the transition (toxic-persona feature or B-vector), the risk that the statistic recovers via unrelated fast modes is substantially reduced. The feasibility stage still verifies that the chosen projection is stable under ordinary training and shows a clear, reversible deflection after a kick; if neither theory-motivated direction satisfies these pragmatic criteria, the best-performing generic statistic is used and the limitation is noted.

Interventional recovery-time protocol

Rationale for weight-space perturbation

Activation-space steering is typically transient. Classical recovery-time measurement (Scheffer et al., 2009; Veraart et al., 2012) requires a persistent displacement that the system's own restoring dynamics must overcome. A one-time additive kick applied to the LoRA weights and left in place satisfies this requirement while continued fine-tuning supplies the restoring dynamics.

We treat the fine-tuning dynamics themselves, gradient descent on the training loss under the fixed poison-fraction mixture, as the dynamical system of interest. The current-fraction equilibrium is the fixed point of that system. A persistent kick displaces the LoRA parameters from this fixed point; resumed fine-tuning (at reduced learning rate) supplies the restoring force. Recovery time is therefore the relaxation time of the optimiser-driven dynamics back toward the unperturbed trajectory. This is a deliberate identification: it is not the free intrinsic relaxation of an autonomous dynamical system in the classical ecology sense (van Nes & Scheffer, 2007; Veraart et al., 2012), but the relaxation of the training process that is actually generating the transition. Because the same optimiser and loss landscape produce both the behavioural change and the recovery, lengthening of this recovery time remains a valid signature of critical slowing down within the dynamical system under study. The reduced learning rate is chosen only to make the return observable; it does not alter the identification.

Kick magnitude and direction

At the single endpoint (step $S_{\max}$ of each (fraction, seed, arm) run, a small additive perturbation is applied to the LoRA parameters and left in place. Because the adapter is rank-1, the learned B-vector spans the one-dimensional update subspace that Turner et al. (2025) showed undergoes abrupt rotation at the mechanistic EM transition. The primary kick direction is therefore chosen to have a controlled component along the current B-vector, maximising the likelihood of displacing the mode relevant to the transition. Magnitude is set to the smallest value that produces a detectable jump in the selected residual-stream statistic, remains reversible under resumed training, and does not itself drive behavioural scores to extreme values. Direction and magnitude are locked during feasibility and then applied uniformly to every fraction and seed.

If a B-vector-aligned kick proves too disruptive in feasibility, a fixed orthogonal or random direction (fixed seed) is the fallback. Recovery time is measured in

reduced-LR optimiser steps.

Post-kick dynamics

After the kick, the kicked model and a parallel unkicked copy continue fine-tuning under identical data order. Learning rate is reduced relative to the main training rate so that recovery is not dominated by large optimisation steps. The reduction factor is chosen in the feasibility stage and frozen. The reduced learning rate is an experimental convenience that slows the optimiser steps so recovery can be resolved; the underlying dynamical system remains gradient descent on the poison-fraction loss.

Recovery time

Recovery time is the number of reduced-LR optimiser steps required for the residual-stream statistic of the kicked run to return to the parallel unkicked trajectory. The operational criterion (distance threshold relative to the initial post-kick distance and the number of consecutive steps (K) that must satisfy it) is locked during feasibility. The identical frozen criterion is applied to every poison fraction and every seed.

Number of recovery measurements

Exactly one recovery-time measurement is obtained per (poison fraction, seed, arm): the measurement is taken at the single endpoint ($S_{\max}$) of that run. This yields one recovery time per point on the poison-fraction axis.

Where compute permits, several independent kicks (different random seeds for the kick direction or magnitude within the locked range) are performed at the same endpoint and averaged; recovery time is known to have high variance, and averaging reduces the noise of each point on the poison-fraction axis.

Parallel trajectories

For every kicked run an unkicked run is maintained under identical data ordering and hyperparameters (apart from the kick). This supplies the baseline against which return is measured.

Feasibility stage

The following quantities are determined on high-poison-fraction checkpoints (and the corresponding benign control) before the full ladder is run, then frozen:

- Exact residual-stream statistic (candidate, layer, aggregation, probe set). Selection criterion: stability under ordinary training + clear, reversible deflection after a kick; preference for a statistic that tracks the direction of change accompanying behavioural EM.
- Kick magnitude and direction (B-vector component as primary plan). Selection criterion: smallest magnitude that is detectable, reversible, and non-transition-inducing.
- Post-kick learning-rate reduction factor. Selection criterion: permits clean return without large stochastic jumps.
- Recovery criterion (distance threshold and consecutive-step parameter K). Selection criterion: neither noise-dominated nor so strict that recovery is never declared.
- Final confirmation of layer index for any non-14B model.
- Fixed step budget $S_{\max}$.

All of the above are frozen after the feasibility stage and applied uniformly.

Experimental sequence and compute

1. One-fraction feasibility check at high poison fraction (and benign control).
2. Lock residual-stream statistic, kick magnitude and direction, learning-rate reduction factor, recovery criterion, and $S_{\max}$.
3. Full coarse poison-fraction ladder $\{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ with >3 seeds per fraction, followed by refined spacing ($\approx 0.05 - 0.1$) around behavioural onset.
4. At the single endpoint (step $S_{\max}$) of each (fraction, seed, arm) run, apply the locked kick protocol and record recovery time (in reduced-LR steps) together with behavioural scores.

Analysis plan

Primary order parameter

The confirmatory test uses the binary EM rate as the sole primary behavioural order parameter. Recovery time of the residual-stream statistic is the

interventional variable whose onset is compared with the behavioural onset. Continuous alignment and any additional internal measures are secondary.

Onset definition and uncertainty

For each seed and each arm, the onset poison fraction of the binary EM rate and the onset poison fraction of recovery-time lengthening are estimated by binary segmentation. Binary segmentation is a formal single-change-point estimator that locates the split in an ordered sequence maximising the improvement in fit when each segment is summarised by its own mean. It is used because the poison-fraction ladder is short and ordered and the scientific question concerns a single transition; with few points the procedure is exact, transparent and computationally trivial. No prior EM study analyses a poison-fraction ladder, so the choice is specific to this pilot.

Per-seed uncertainty is obtained by bootstrap resampling of the responses (or residual-stream trajectories) at each fraction and re-estimating the change point, yielding a percentile confidence interval on the onset fraction. The identical estimator is applied to both the behavioural series and the recovery-time series.

Primary test

For each seed compute the paired difference

$$\Delta = \text{behavioural-onset fraction} - \text{recovery-onset fraction}$$

A positive Δ means recovery lengthening appears at a lower poison fraction than the behavioural rise.

With three seeds, any interval estimate is wide and its coverage only approximate; the bootstrap rule is therefore treated as a pilot-scale effect estimate rather than a definitive confirmatory test, and at least five seeds are used wherever the 14B compute budget permits.

The **primary decision rule** is a bootstrap confidence interval for the mean Δ : the interval must lie entirely above 0. This rule is well-defined with >3 seeds and does not require the larger sample sizes needed for a sign test to have adequate power. A one-sided sign test against the null $\Delta \leq 0$ is reported as a secondary robustness check.

On the benign-capability control no behavioural EM onset exists, so Δ is undefined there; instead we estimate only the recovery-onset fraction and its CI along the benign ladder. A positive overall result requires:

- A bootstrap CI for mean Δ entirely above 0 on the harmful arm.
- Recovery-time lengthening on the harmful arm that is absent or significantly weakened on the benign arm.

Resolution floor

If the true difference in onsets is smaller than the refined ladder spacing, the ordering cannot be resolved. This is an explicit limitation of the design and is the reason refined spacing is used around the behavioural onset region. Differences smaller than the local spacing are reported as unresolved.

In light of the rapid transition reported by Hennick & Corlouer (approximately 20% alignment drop within 5–10 steps), both the behavioural onset and the recovery-time onset may fall inside the same narrow fraction window; the design therefore treats a null or unresolved Δ as a plausible and informative outcome rather than a failure of the protocol.

Null and mixed outcomes

No earlier lengthening (or failure of the primary decision rule) is interpreted as evidence against, or only weak evidence for, classical critical slowing down under this protocol. Concurrent lengthening or lengthening that also appears on the benign control is recorded as such.

Scoped claims and exclusions

A positive result is interpreted as evidence consistent with an approaching critical transition. The pilot does not claim that the transition is a fold bifurcation, does not test bistability via hysteresis, does not scale to frontier models, does not directly compare recovery time with spectral statistics, and does not introduce additional datasets. These extensions remain future work.

Results

Discussion

Conclusion

References

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